Md Omar Faruqe Riyad

Software Engineer | Backend Systems & AI Infrastructure

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Education

Shahjalal University of Science and Technology (SUST) — Sylhet, Bangladesh
B.Sc. in Computer Science and Engineering; CGPA: 3.53/4.00 (Last two years: 3.83) — *2018 – 2023*

Experience

Dynamic Solution Innovators — April 2023 – Present
Software Engineer — *Dhaka, Bangladesh*

- **EC-BVRS:** Contributed to the data pipeline behind the voter registration flow of Bangladesh's national voter registration (NID) platform, which processes records for nearly **150 million voters** (Spring Boot, Kafka, Redis).
- **Tooling:** Built and open-sourced `gf`, a GlassFish CLI that hot-swaps modified bytecode into running JVMs over JDI/JDWP and bypasses JasperReports classloader caching. Cut local edit-deploy cycles from **∼2 minutes to ∼5 seconds**.
- **E-Appeal (NBR):** Consolidated authorization checks scattered across JSF views and service code into a single permission-evaluation layer for role-based access control.
- **SHMS:** Built a configurable approval-workflow engine that replaced hardcoded per-module approval logic. Added a field-level correction flow so reviewers can reopen individual form fields with inline feedback instead of rejecting whole submissions.
- **SHMS:** Introduced Prometheus, Loki, and Grafana observability for GlassFish applications, exposing JVM memory, thread pools, and request latency for production debugging and capacity planning.
- **BCIC-ERP:** Replaced base64-inlined file rendering with URL-based streaming behind a pluggable `StorageClient` interface (MinIO or filesystem), cutting page payload size and server memory use. Files are served through nginx X-Accel-Redirect after per-request authorization checks.
- **BCIC-ERP:** Moved SMS and email notifications onto a JMS queue with dead-letter retries, so provider outages no longer block user-facing requests. Led development of the Procurement, Inventory, Asset, and Budget modules (4 of 12), from process modeling and schema design through production rollout.
- **Cross-project:** Isolated auditing and notification dispatch from business logic with AspectJ cross-cutting interceptors, a pattern reused in multiple Jakarta EE systems.

Altri.ai — Part-time
Full-Stack AI Engineer, RevestAI — *Remote, US*

- Rewrote the sequential LLM enrichment pipeline as bounded-concurrency async batches with atomic per-batch commits. Throughput rose about **10×**, and failed runs resume without repeating completed LLM calls.
- Designed the three-stage property scoring pipeline that separates LLM feature extraction, ML valuation, and deterministic financial heuristics, so each layer can be retrained and tested on its own.
- Built the property ingestion pipeline on HomeHarvest and PostgreSQL with content-hash deduplication and incremental delta-syncing. Repeated scrapes no longer trigger duplicate enrichment runs, and historical listing state is preserved.
- Implemented distributed job coordination on PostgreSQL with `SELECT FOR UPDATE` row locks and stale-lock recovery, so multiple scheduler replicas safely share long-running enrichment and training jobs.
- Built the training and promotion pipeline for XGBoost property valuation. A challenger must beat the champion on a gold-standard holdout slice, and a human signs off before production-wide rescoring, since aggregate metrics can hide slice regressions and valuation errors cost money.

Projects

SREGym Fault Scenario Contribution | *Python, Kubernetes, Docker* — 2026

- Reproduced a documented Kubernetes admission-control incident (issue #128162) where cumulative mutating-webhook timeouts exceed the API server's admission deadline and silently block all pod creation. Merged into SREGym, an open-source benchmark for AI SRE agents.
- Designed a four-property mitigation oracle with a fresh probe-pod admission test that rejects reward-hacking fixes, such as deleting all webhooks, that mask the symptom without restoring the control-plane network path.

Token-Efficient LLM Query Router | *AMD Hackathon Act II, Track 1* — 2026

- Designed a four-tier inference cascade (regex router → ONNX classifier → LoRA fine-tuned Qwen2.5-1.5B → cloud LLM fallback) that resolves 8 task categories with minimal paid API usage, admitting each tier only at measured precision (router 64/64 on adversarial cases; sentiment classifier 100%).
- Replaced model confidence checks with deterministic verification guards: sandboxed AST evaluation of model-emitted math expressions, which removes arithmetic errors and cuts output tokens ∼90%, and an entity-completeness check that catches 99% of dropped entities in NER.
- Fine-tuned the local tier on 2,800 programmatically validated synthetic examples and iterated 17 times against an LLM-as-judge harness mirroring the competition rubric, root-causing each regression before redeploy.

KnowledgeRelay | *Python, FastAPI, LangChain, ChromaDB, React* 2025

- Built a knowledge-transfer agent that interviews outgoing team members with adaptive questions, ingests their documents and code, and answers new joiners' questions with source attribution. Retrieval rewrites follow-ups into self-contained queries.

PrimeFaces Open Source Contribution | *Java, JSF, PrimeFaces* 2024

- Contributed a feature enhancement to PrimeFaces' `JPALazyDataModel` that lets developers inject custom filters into DataTable-backed JPA queries.

Research & Publications

AI for Site Reliability Engineering (SREGym) | *Research Intern, advised by Prof. Tianyin Xu* UIUC, 2026

- Working on the SREGym v2 paper: extending the benchmark from synthetic faults to real-world failure scenarios and analyzing how agents behave on them.

AdversaryAI at BLP-2025 Task 2 BLP 2025, IJCNLP-AACL

- Improved low-resource Bangla code generation with LoRA fine-tuning and an execution-guided self-refinement loop (Pass@1 0.85).

Deja Vu at SemEval 2024 Task 9 SemEval 2024, NAACL

- Designed data augmentation strategies and compared fine-tuned and prompted language models on lateral-thinking puzzles.

Team_Syrax at BLP-2023 Task 1 BLP 2023, EMNLP

- Built a semi-supervised pipeline with self-training, back-translation, and ensembling for Bangla text classification.

Visual Question Answering (VQA) Data Synthesis | *Undergraduate Thesis* 2022

- Generated diverse, context-aware question-answer pairs to widen answer-space coverage in VQA training data.

Honors, Awards & Certifications

4th place out of 32 teams, Code Generation in Bangla Shared Task, BLP at IJCNLP-AACL 2025 [Leaderboard]

6th place out of 30 teams, DSI AI Agent Hackathon 2025 [Leaderboard]

12th place in post-evaluation (19th official), Violence Inciting Text Detection, BLP at EMNLP 2023 [Leaderboard]

11th place, BUET CSE Fest 2022 AI Contest

Certified in Cybersecurity (CC), ISC2 [Badge]

Teaching

Course Instructor 2024 – Present
Dynamic Learning *Remote*

- Authored and taught Enterprise Web Development with JEE, JSF, and PrimeFaces: component-driven design, CDI/EJB integration, and JSF lifecycle management.

Technical Skills

Languages: Java, Python, SQL, Bash, JavaScript, C/C++
Backend & Systems: Jakarta EE (JSF, CDI, EJB, JPA, JMS), Spring Boot, FastAPI, Kafka, GlassFish, AspectJ
AI & ML Systems: PyTorch, LLMs, RAG, LangChain, ChromaDB, XGBoost, LoRA fine-tuning
Databases & Storage: PostgreSQL, MySQL, Redis, Supabase, MinIO, Hibernate/JPA, Flyway, Alembic
Security: RBAC, Jakarta Security, JWT
Infrastructure & Observability: Docker, Kubernetes, Git, GitHub Actions, Prometheus, Grafana, Loki, Nginx, Ansible
Frontend: React, PrimeFaces